

Inbreeding

Biologia della Conservazione 25/26

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- ▶ Inbreeding is defined as the mating between related individuals
- ▶ When two related individuals produce an offspring, that individual can receive two alleles that are identical by descent, i.e. they can be homozygous by descent (sometimes termed autozygous)

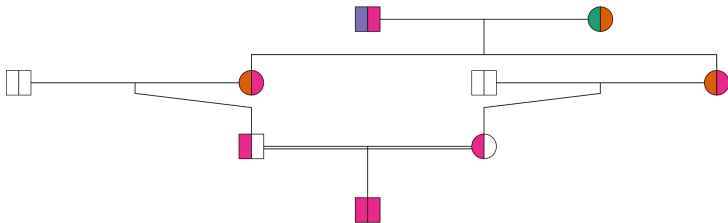


Figure 1: Mating between first cousins

- ▶ Can inbreeding happen when individuals are mating at random?

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- ▶ Imagine the case of small populations. In the case of a population with 10 breeders, after a few generations the most distantly related individuals will be cousins, and even if the population mates at random, individuals will be highly inbred
- ▶ This has been called inbreeding effect of small populations (Crow & Kimura, 1970)
- ▶ So, we can consider an individual inbred if its parents share a common ancestor and its genes (one, two, or big chunks of the genome) are Identical by Descent (IBD)

- ▶ What we know from studying evolution and from the examples on the coalescence theory is that eventually all parents of an individual share a common ancestor
- ▶ We must, therefore, define inbreeding relative to some “base” population in which we assume all individuals are unrelated to one another (Allendorf et al., 2022)
- ▶ The concept of inbreeding is relatively easy to understand, but can also be confusing as the same term is associated with different phenomena (genetic drift, non random mating within populations, increase in genome wide homozygosity)

- ▶ All these concepts are measured by the same quantity, inbreeding coefficients (Templeton & Read, 1994), and all these concepts can, to certain degree, be used interchangeably as they all refer to a situation in which relatives mate with each other (Keller & Waller, 2002)
- ▶ As inbred individuals receive two copies of the same allele from its parents, they will present higher homozygosity and lower heterozygosity throughout the entire genome
- ▶ We can, then, define the inbreeding coefficient (F) as that proportion of the genome in an individual that is IBD (sometime an alternative word is used, i.e., autozygous)
- ▶ The alternative to IBD is IBS (Identical by State), which means that an individual possesses two alleles that are descendant from different ancestral alleles in the base population

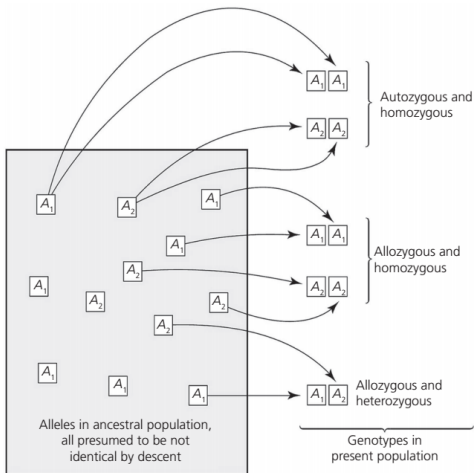


Figure 17.2 Patterns and definitions of genotypic relationships with pedigree inbreeding. Autozygous individuals in the present population contain two alleles that are IBD from a single gene in the ancestral population. In contrast, allozygous individuals contain two alleles derived from different genes in the ancestral population. Redrawn from [Hartl & Clark \(1997\)](#).

Figure 2: Figure from Allendorf et al. (2022), pg. 374

The pedigree inbreeding coefficient

- ▶ The pedigree inbreeding coefficient (F_P) refers to the expected increase in homozygosity of the inbred individual relative to the population founders or founder
- ▶ The value ranges from 0 (indicating no inbreeding) to 1 (indicating completely inbred)

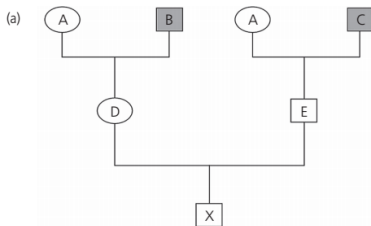


Figure 3: Classical representation of a pedigree. Squares are males and rounds are females. After Allendorf et al. (2022)

(b)

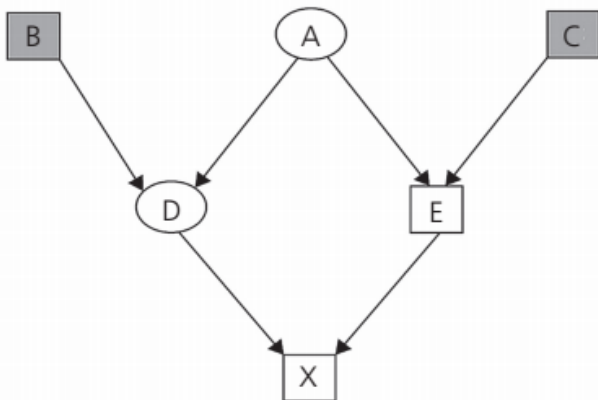


Figure 4: Path diagram representing the pedigree. Approach used to calculate the inbreeding coefficient F_P

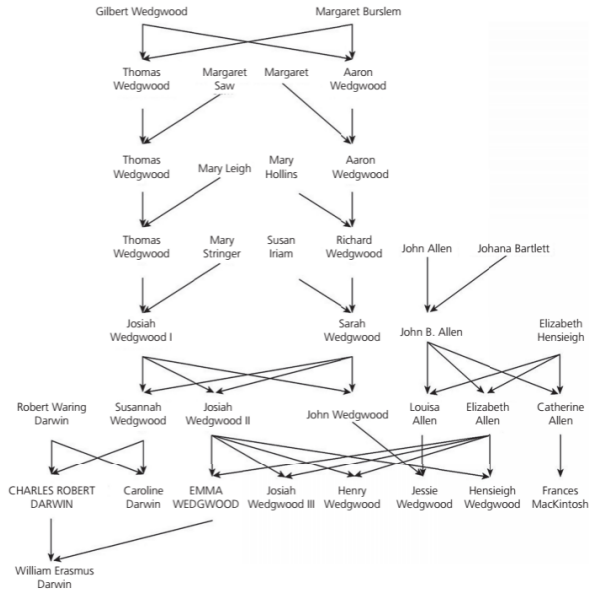


Figure 5: Did you know that Charles Darwin and his wife were cousins?

How to estimate pedigree?

1. Draw the pedigree as shown in previous figure where each individual appears only one
2. Identify the ancestors of the male and female that generated X. If the father and the mother share no ancestors $F_P = 0$
3. Trace all of the paths that lead from one of X's parents, through the common ancestor, and then back again to the other parent of X (in our example there is only one path: from D to A and from A to E. The notation to indicate such path is DAE)
4. Determine N , the number of individuals in the path, excluding our target individual X. In our case $N = 3$

5. With a single path we can use

$$F_P = \left(\frac{1}{2}\right)^N (1 + F_{P(CA)}) \quad (9.1)$$

with $F_{P(CA)}$ is the pedigree inbreeding coefficient of the common ancestor

6. If there is more than just one path, F_P becomes the sum of all F_P values obtained from each single paths

$$F_P = \sum \left[\left(\frac{1}{2}\right)^N (1 + F_{P(CA)}) \right] \quad (9.2)$$

- So, in our previous example, taken from Allendorf et al. (2022), we can use equation 7.1 and the estimated inbreeding coefficient is

$$F_{P(X)} = \left(\frac{1}{2}\right)^3(1 + 0) = 0.125$$

- We can read this result in two complementary ways:
1. Individual X is expected to have 12.5% of his genes IBD
 2. X is expected to have its heterozygosity reduced by 12.5% compared to the ancestral population

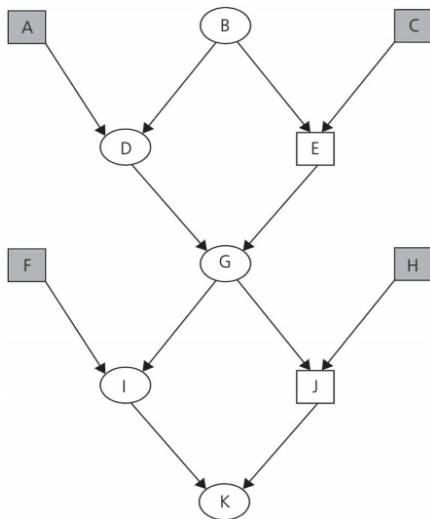


Figure 6: What is $F_{P(K)}$ based on the above path diagram representation?
After Allendorf et al. (2022)

- ▶ There is one loop that contains a common ancestor of both parents of K (IGJ). Therefore, using equation $F_{P(K)} = (\frac{1}{2})^3(1 + F_{P(G)})$
- ▶ Individual G is also inbred; there is one loop with three individuals through a common ancestor for individual G (DBE). Therefore, $F_{P(K)} = (\frac{1}{2})^3(1 + F_{P(B)})$. Individual B is not inbred ($F_{P(K)} = 0$) since she is a founder in this pedigree
- ▶ Solving the equations: $F_{P(G)} = 0.125$, and $F_{P(K)} = (\frac{1}{2})^3(1.125)$

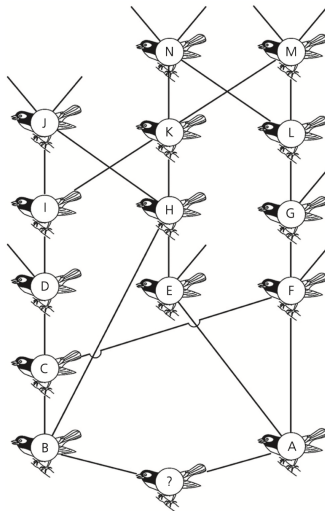


Figure 7: Complex pedigree. After Allendorf et al. (2022)

Path	Length (N)	Common Ancestor	$F_{P(CA)}$	$(1/2)^N(1+F_{P(CA)})$
BC <u>F</u> A	4	F	0	0.0625
B <u>H</u> E A	4	H	0	0.0625
BCDI <u>K</u> HEA	8	K	0	0.0039
BCDI <u>J</u> HEA	8	J	0	0.0039
BHK <u>M</u> LGFA	8	M	0	0.0039
BHK <u>N</u> LGFA	8	N	0	0.0039
BCDIK <u>N</u> LGFA	10	N	0	0.0010
BCDIK <u>M</u> LGFA	10	M	0	0.0010
BCFGL <u>M</u> KHEA	10	M	0	0.0010
BCFGL <u>N</u> KHEA	10	N	0	0.0010
Total	—	—	—	0.1445

Figure 8: Calculation of the pedigree inbreeding coefficient for the previous slide. After Allendorf et al. (2022)

- ▶ Two major short comings while using F_P :
 1. It does not account for the inbreeding level of individuals that are not included in the reference pedigree
 2. This metric is sub-optimal to obtain an accurate estimate of the proportion of the genome that is IBD, because there is a much variation in the proportion of genes that are IBD imputable to chance alone
- ▶ In general, it is often very complicated to obtain accurate pedigrees for natural populations, and while this approach is still massively used in all species for which we can maintain appropriate pedigree databases, coefficient of inbreeding in wild animals is often obtained through the analysis of molecular markers

- ▶ Molecular marker-based measures of F range from simple estimates of individual heterozygosity to more advanced methods that use mapped loci to estimate F via identification of IBD chromosome segments as stretches of homozygous genotypes at mapped single nucleotide polymorphisms (Allendorf et al., 2022, pg. 381)

- ▶ Estimates of F based on individual heterozygosity rely on the concept that individuals whose parents are more closely related will have lower heterozygosity on average across the genome due to the presence of IBD chromosome segments

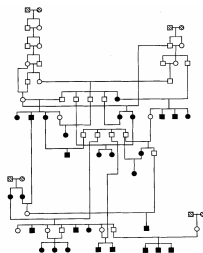


Fig. 1. Pedigree of the captive Swedish wolf population. Only the parts of the pedigree leading to the animals analysed in this study (filled symbols) are shown. Founders are indicated by shaded symbols.

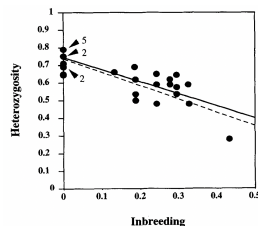


Fig. 2. The relationship between individual heterozygosity (H_i) at microsatellite loci and inbreeding (f) in a captive wolf population. The straight line represents the regression of H_i on f , the dashed line is the expected relationship between H_i and f , assuming an H_i of 0.75 among unrelated individuals.

Figure 9: After Ellegren (2004)

Figure 10: After Ellegren (2004)

- ▶ The availability of thousands of genetic markers or more now makes it possible to use this approach in many taxa
- ▶ F_H measures the excess in the observed number of homozygous genotypes within an individual relative to the mean number of homozygous genotypes expected under random mating

$$F_H = \frac{O(Hom) - E(Hom)}{L - E(Hom)} \quad (9.3)$$

- ▶ Where $O(Hom)$ and $E(Hom)$ are the observed and expected Hardy-Weinberg (HW) numbers of homozygous loci in an individual and L is the number of loci examined
- ▶ The average $O(Hom)$ is expected to equal $E(Hom)$ in random mating populations. F_H can thus be negative when the parents are less related than expected with random mating

Inbreeding coefficient using mapped loci

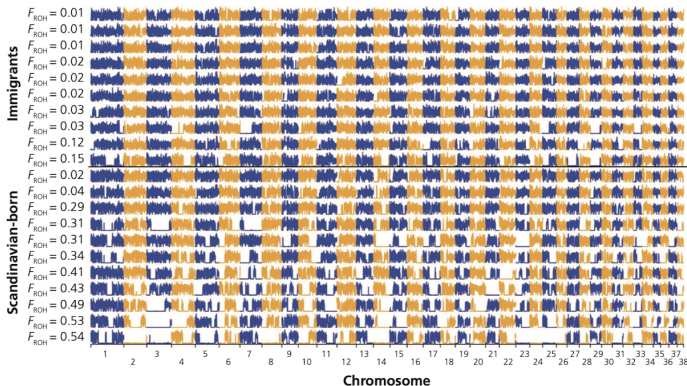


Figure 11: Heterozygosity across chromosomes 10-14 of Scandinavian Wolves as analyzed by Kardos et al. (2017).

- ▶ Inbred individuals generally have reduced fitness in comparison to non-inbred individuals from the same population because of their increased homozygosity
- ▶ Inbreeding depression is the reduction in fitness (or phenotype value) of progeny from matings between related individuals relative to the fitness of progeny between unrelated individuals
- ▶ Inbreeding depression in natural populations contributes to the extinction of populations under some circumstances (Allendorf et al., 2022)
- ▶ The importance of inbreeding depression has been debated since the time of Darwin. Even today, the importance of inbreeding depression is sometimes questioned by some conservationists and policy-makers

A common garden example

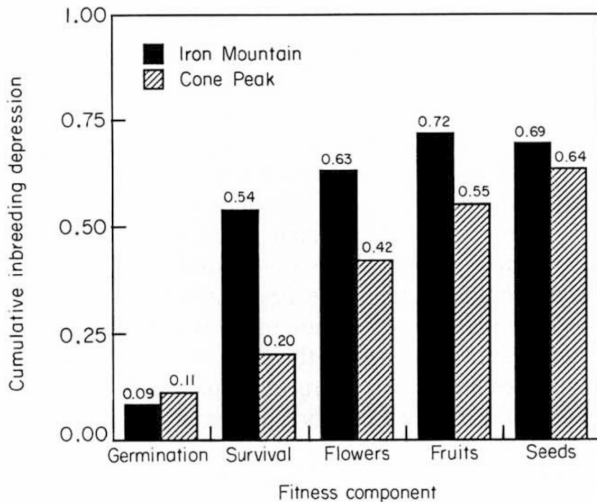


Figure 12: Common garden experiment of *Mimulus guttatus*

An example from a zoo population



Figure 13: Callimico monkeys from the **Connecticut Zoo**

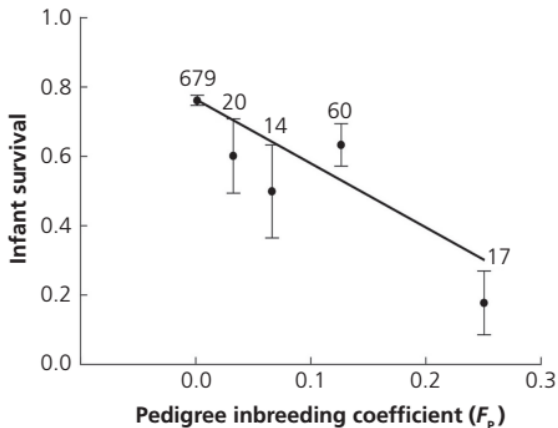


Figure 14: Relationship between inbreeding (F_P) and infant survival in captive callimico monkeys. After Lacy (1997)

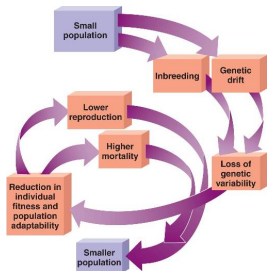


Figure 15: The concomitant contribution of different factors in determining a vortex effect leading the populations to extinction. Image downloaded [here](#)

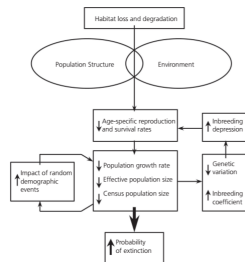


Figure 16: A more recent take on extinction vortex. After Allendorf et al. (2022)

- ▶ Inbreeding depression can result from either increased homozygosity or reduced heterozygosity
- ▶ Aren't increased homozygosity or reduced heterozygosity two sides of the same coin?
- ▶ Yes, but the mechanism related to a fitness reduction is distinct! Increased homozygosity leads to expression of a greater number of deleterious recessive alleles in inbred individuals, thereby lowering their fitness. Reduced heterozygosity reduces fitness of inbred individuals at loci where the heterozygotes have a selective advantage over homozygotes (heterozygous advantage or overdominance)
- ▶ Most evidence show support for the primary problem of inbreeding depression is increased homozygosity (Allendorf et al., 2022)

Different mechanisms of fitness reduction

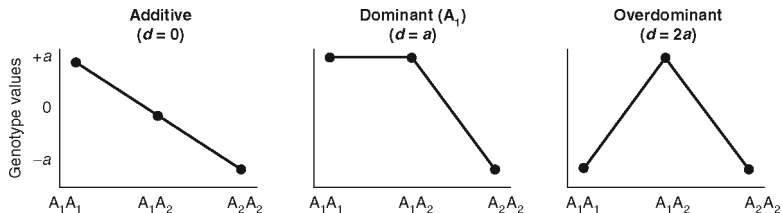


Figure 17: After Frankham, Ballou, & Briscoe (2010)

- ▶ It is crucial to know the mechanisms causing inbreeding depression because it affects the ability of a population to “adapt” to inbreeding
- ▶ A population could adapt to inbreeding if inbreeding depression is caused by deleterious recessive alleles that potentially could be removed (purged by selection)

- ▶ However, inbreeding depression caused by heterozygous advantage cannot be purged because overdominant loci will always suffer reduced fitness as homozygosity increases due to increased inbreeding (Allendorf et al., 2022)
- ▶ Deleterious recessive alleles are present in the genome of all species because they are continually introduced by mutation, and natural selection is inefficient at removing them because most copies are “hidden” phenotypically in heterozygotes that do not have reduced fitness (Allendorf et al., 2022)
- ▶ We, therefore, expect all species to show some inbreeding depression due to the increase in homozygosity of recessive deleterious alleles

- ▶ The effects of inbreeding depression on survival are often measured by the mean number of lethal equivalents (LEs) per diploid genome
- ▶ A lethal equivalent is a set of deleterious alleles that would cause death if homozygous
- ▶ One lethal equivalent may be a single allele that is lethal when homozygous, two alleles each with a probability of 0.5 of causing death when homozygous, or 10 alleles each with a probability of 0.10 of causing death when homozygous

- ▶ The number of LEs present in a species or population is generally estimated by regressing survival on the inbreeding coefficient
- ▶ The effects of inbreeding on the probability of survival, S , can be expressed as a function of F_P following Morton, Crow, & Muller (1956)

$$S = e^{-(A+BF_P)} \quad (9.4)$$

where e^{-A} is survival in an outbred population, and B is the inbreeding load.

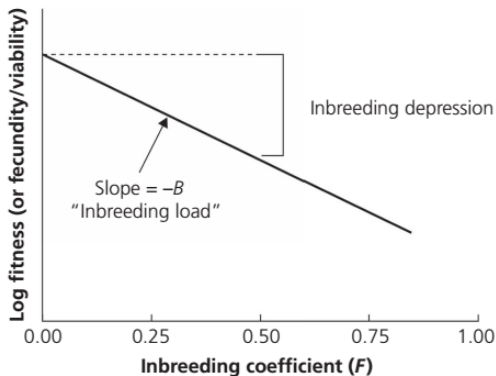


Figure 18: Relationship between inbreeding coefficient (F) and reduction in fitness. Inbreeding depression is the reduction in fitness of inbred individuals and measured by the number of LEs per gamete (B). The bracket shows the reduction in fitness (inbreeding depression) predicted for individuals with an F of 0.5. After Keller & Waller (2002) and Allendorf et al. (2022).

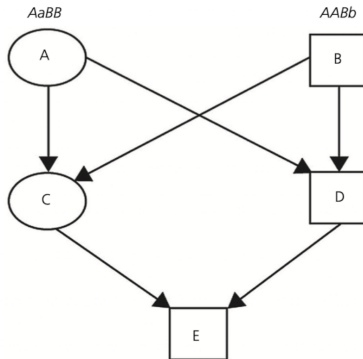


Figure 17.13 Effect of a single lethal equivalent on survival of inbred progeny produced by a mating between full sibs ($F_P = 0.25$). Individuals A and B each carry one lethal allele (*a* and *b*, respectively). The probability of E being homozygous for the *a* allele is $(1/2)^4 = 1/16$; similarly, there is a $1/16$ probability that individual E will be homozygous *bb*. Therefore, the probability of E *not* being homozygous for a recessive allele at either of these two loci is $(15/16)(15/16) = 0.879$. This demonstrates that 1 LE per diploid genome will result in approximately a 12% reduction ($1 - 0.879$) in survival of individuals with an F_P of 0.25.

Figure 19: After Allendorf et al. (2022)

- ▶ Genetic load is the relative difference in fitness between the theoretically fittest genotype within a population and the average genotype in that population.
- ▶ The term load was first used by Muller (1950) in his consideration of the possible effects of increased mutation rates because of nuclear weapons
 1. Mutation load: The decrease in fitness caused by the accumulation of deleterious mutations
 2. Segregation load: The decrease in fitness caused by heterozygous advantage
 3. Drift load: The decrease in fitness caused by the increase in frequency of deleterious alleles resulting from genetic drift
 4. Migration load: The reduction in fitness caused by the migration into a population of individuals that are less adapted to the local environment than native individuals

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